MODERATED LEARNING MANAGEMENT SYSTEM (LMS)

Link To Git Repository: https://github.com/Kevin-droid408/CS4433-Moderated-Learning-Management-System

BUILD GUIDE

1. Development Environment

The Moderated LMS is developed using the following tools and technologies:

- **Programming Language:** C++17
- **IDE:** Visual Studio Code
- **Compiler:** GNU C++ (g++) through MSYS2
- **Build System:** CMake
- **Database:** SQLite
- **Build Tool:** MinGW Make

2. Project Configuration

The project uses CMakeLists.txt to define the build configuration, including the C++ standard, source files, include directories, and SQLite library dependencies.

The project requires C++17 and SQLite to compile successfully.

3. Build Instructions

Open the Moderated LMS project folder in Visual Studio Code and open the integrated terminal.

Step 1: Verify CMake

cmake --version

Step 2: Verify the C++ compiler

g++ --version

Step 3: Configure the project

cmake -S . -B build -G "MinGW Makefiles" -DCMAKE_MAKE_PROGRAM="C:/msys64/ucrt64/bin/mingw32-make.exe"

Step 4: Compile the application

cmake --build

If the build completes successfully, the executable should be generated in the build directory.

4. Running the Application

Run the following command from the project root directory:

.\build\ModeratedLMS.exe

5. Build Verification

The build is considered successful when:

- CMake configures the project without errors.
- All required C++ source files compile successfully.
- SQLite links successfully.
- The executable is generated.
- The application launches successfully.